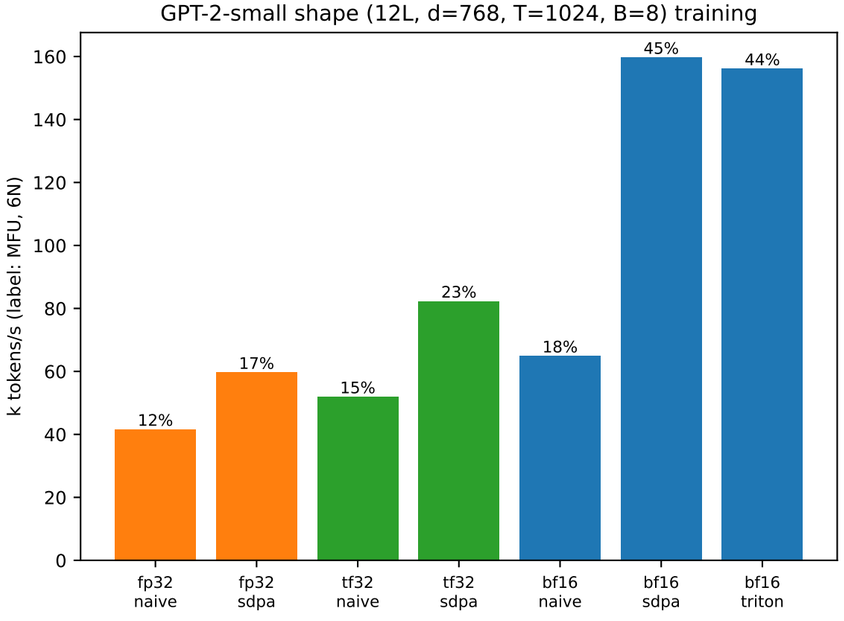
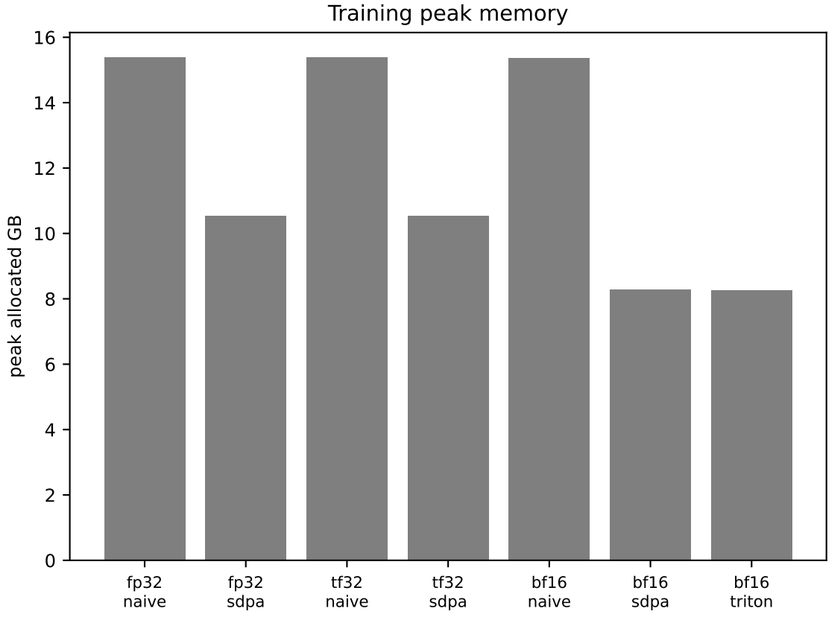
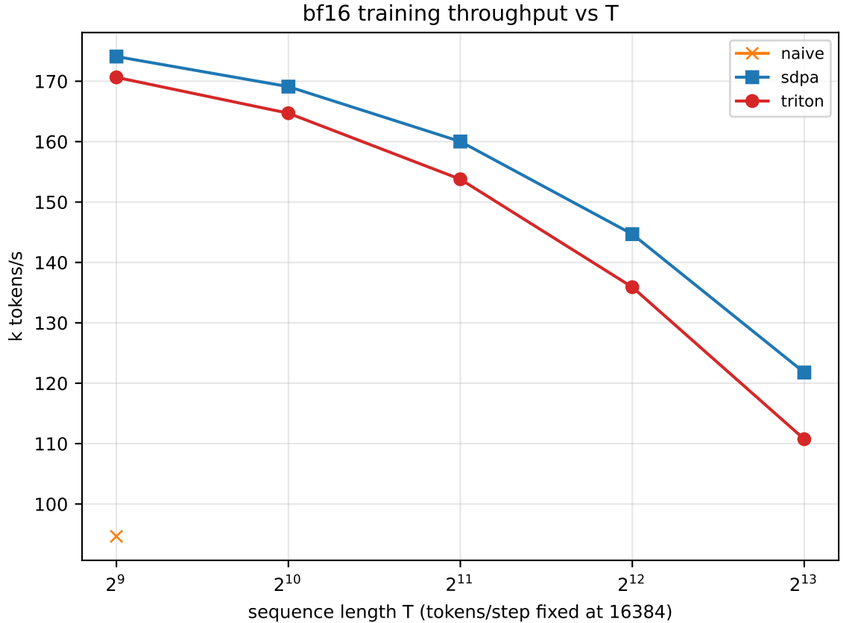

GPT-2-small shape (12L, d=768, T=1024, B=8) training
k tokens/s (label: MFU, 6N)
0
20
40
60
80
100
120
140
160
12%
17%
15%
23%
18%
45%
44%
fp32 naive
fp32 sdpa
tf32 naive
tf32 sdpa
bf16 naive
bf16 sdpa
bf16 triton


Training peak memory
peak allocated GB
0
2
4
6
8
10
12
14
16
fp32 naive
fp32 sdpa
tf32 naive
tf32 sdpa
bf16 naive
bf16 sdpa
bf16 triton


bf16 training throughput vs T
k tokens/s
100
110
120
130
140
150
160
170
naive
sdpa
triton
2^9
2^{10}
2^{11}
2^{12}
2^{13}
sequence length T (tokens/step fixed at 16384)